Candidate Seat no…………

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech (EC) Examination

APRIL-MAY 2018

EC 247 Microprocessor & Interfacing

Date: 02.05.2018, Wednesday **Time: 10:00 a.m. To 01.00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q-1 **Do as Directed.** **07**

(a) The instructions that are used to call a subroutine from a main program and return to the main program after execution of called function are **01**

a) CALL,JMP b) JMP,IRET

c) CALL,RET d) JMP,RET

(b) NOP instruction introduces **01**

a) Address b) Delay

c) Memory location d) None of the mentioned

(c) The Intel 8086 microprocessor is a_____ processor. **01**

a) 8 bits b) 4 bits

c) 16 bits d) 32 bits

(d) What is /are the improvement is in the architecture of 8086 over 8085 architecture **01**

a) A 16-bit ALU, a set of 16-bit registers

b) Segmented memory addressing

c) Fetched instruction queue for overlapped fetching and execution

d) All of the mentioned

(e) The length of pre-decoding instruction byte queue is _______ bytes long. **01**

a) 2 b) 4

c) 6 d) 8

(f) If segment address = 1005 H, offset address = 5555 H, then the physical address is_____. **01**

a) 655A H b) 155A5 H

c) 4550 H d) 56555

(g) When one segment starts before the end of another segment then we call them as_______. **01**

a) Non-overlapping segments b) Overlapping segments

c) Stack area d) None of these

Q-2 (a) Interface following memories with 8086. **07**

128KB of ROM starting from 00000H.

128KB of ROM starting from 40000H.

128KB of RAM starting from 80000H.

(b)	Interface 8-LEDs with 8086 using 8255 and Write an assembly language code to turn on and off all LEDs after every 500ms.	**07**
	OR	
(b)	List all steps for interrupt execution in 8086.	**07**
Q-3 (a)	Interface 2 kb RAM and 1kb ROM with 8085 processor to satisfy the total memory requirement of 4 kb RAM (0000h-0FFFH) and 2kb ROM (8000h-87FF).	**07**
(b)	Differentiate between 8085 and 8086 Microprocessors.	**04**
	OR	
(b)	Draw timing diagram of instruction fetch cycle(assume 4 states machine cycle).	**04**
(c)	Discuss the importance of ALE pin in Microprocessor.	**03**

SECTION – II

Q-4	Interface 8255 programmable peripheral interface with 8086.(Assume PORTA address of 8255 is B0h). (a) Write a control word to initialize PORTC as input and PORTA and PORTB as output in Mode 1. (b) Write a control word to set PC5.	**07**
Q-5(a)	Interface 2-common cathode 7 segment with 8086 using 8255 and Write an assembly language code to display 00 to 99 up counter.	**07**
(b)	Interface 8254 programmable timer/counter with 8086 and Write an assembly language code to generate 20 KHz and 100Khz square wave simultaneously on counter 0 and counter 1.(Assume input clock Frequency to 8253 is 2 MHz).	**07**
	OR	
(b)	Interface 16*2 Alphanumeric LCD with 8086 using 8255 and Write an assembly language code to display “SUMMER 2018” on LCD.	**07**
Q-6(a)	Write an assembly language code to exchange block of 10 data bytes from 20000H and 21000H (Assume DS:2000H).	**07**
(b)	An 8086 based system has the following memory requirements: 32KB of ROM starting from 20000H. 64KB of RAM starting from A0000H. Chips available: 2 chips of 16KB ROM , 2 chips of 32KB RAM , 1 chip of 74LS138 3 to 8 decoder. Design the memory interfacing circuit.	**07**

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